FACULTY OF ENGINEERING AND ARCHITECTURE

DETOUR

Technical report

Course: Design Thinking

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1 Introdution

this is the introduction

2 Design thinking phases

Our design thinking phase was far from a linear process. Some phases were passes through multiple times/revisited. In the following sections, the different phases will be discussed as one complete phase, which for us doesn't mean that all the described action, decisions etc were made in the same week/period.

2.1 Empathize phase

2.2 Define phase

2.3 Ideate phase

2.4 Prototype

2.5 Test

3 State of the Art

There are already hundreds of smoking cessation apps available, but unfortunately, most apps are not scientifically validated [3]. In a study by Bold et al., only 11 apps had published results supporting their efficacy.

3.1 Feasibility of apps

While many apps are available, the effectiveness of these apps may be limited [1], [5]. Luckily, it has been shown that digital interventions, through apps, can work [5], [7], when the right interventions are used. Even people who already make use of face-to-face care can find benefits in using digital help [5].

3.2 Description of existing apps

From the literature, we see that a lot of apps do not use science-based interventions [4].

To collect information about the most popular smoking cessation apps, we searched the internet for the top-rated apps, and looked at Facebook groups to see what users recommend to each other.

We looked at a few of the most popular smoking cessation apps, and we saw that they often employ these methods:

- **Progress trackers.** There are many variations of this; some track the number of days without smoking, some track the amount of money saved by not smoking, some track lifetime saved by not smoking.
- **Social aspects.** Some apps use a social aspect, like forums or other chat functions.
- **Planners.** Apps might help users plan designated smoke-free days, or plan a quit date.
- **Cognitive behavioral therapy (CBT)**. Some apps use CBT as a basis for their interventions.
- **Gamification.** Many apps use badges, mileposts, or levels to signify progress. Some apps also include games as a distraction for when a craving hits.

To make things more concrete, we will now look at a subset of the most popular apps, of which we believe they are scientifically based.

3.2.1 QuitSure

The authors of this app have published a study showing its effectiveness [2], [4]. The study reports that 80.1% of the users maintained abstinence for at least 30 days after program completion [4]. Note that this is only for users who have completed the entire program, and does not include users who dropped out, nor the people who did not respond to the follow-up questions. Also note that these are self-reported figures, and have not been biochemically verified.

The app is based on techniques from psychology, CBT, and mindfulness [4].

In an onboarding process, the user is asked to think about their smoking quantity, patterns, past quitting attempts, and relapses. The app then assesses if the user is ready to quit. After onboarding, the user is intended to follow a program consisting of content and video exercises. This program requires 6 to 10 hours over 6 to 12 days to complete [4].

In their discussion, the researchers note that there is a high dropout rate of 59.4% [4], as a result of the high engagement requirement. Additionally, most relapses were a result of high cravings, for which the app did not have specific interventions after program completion.

Unfortunately, the study does not go into detail on what specific modules in the app contributed most to successful smoking cessation.

3.2.2 Quit Sense

While not yet released to the public, there has been a pilot study that shows an 11.5% biochemically verified quit rate after 6 months [6]. This number does include the people who dropped out.

This app is a just-in-time adaptive intervention (JITAI) [6]; it tracks the users location and learns about the environment and location where the user smokes. When a user is in such an

environment, the app sends in-the-moment advise to help them resist smoking.

Upon first opening the app, the user needs to set their quit date. Then, they need to submit their smoking habits and key environmental factors in real-time, via a behaviour logging tool. After the quit date, the app uses location and the previously self-identified high-risk locations to tailor the timing and contents of the support messages [6].

The app was found to be more effective than the control group that used the SmokeFree service [6] provided by the National Health Service (UK) (11.5% vs 2.9%).

The study did not find why the app worked [6], or what specific interventions helped most. Many users cited learning about their triggers and habits as being very valuable. The main reasons for dropping the app [6] were relapse, feeling like they no longer needed support, finding that the app did not match their needs, and technical issues.

3.2.3 Discussion

Both of these apps had studies showed their effectiveness in smoking cessation. However, these apps have one "expected path" to follow, and users drop the app if it doesn't fit their needs. The studies did not find why their app worked, or what specific interventions helped most.

3.3 Behavior models

Through our first meeting with Geert Crombez, we already came into contant with the Health Action Process Approach (HAPA) and Behavioral Change Techniques (BCT) models. In our own search, we came across the Capability-Opportunity-Motivation-Behavior (COM-B) and Theoretical Domains Framework (TDF) models.

These models all serve a slightly different purpose and operate on different timescales.

The COM-B model [citation needed] describes the relationship between capability, opportunity, and motivation in influencing behavior. This means you can influence behavior by tweaking the other factors. The model can thus describe what factors you need to change.

HAPA [8], [9] makes a distinction between a preintentional phase and a volitional phase. In the first phase, a person develops their motivation and plan of action. Once this "good intention" is sufficiently developed, the desired health action can be initiated. Now, in the volitional phase, this action needs to be maintained, which needs self-regulatory skills and strategies [8].

Compared to COM-B, HAPA focuses more on the psychological processes that drive behavior change, particularly the transition from intention to action and the maintenance of that action, rather than the physical or environmental factors.

We want to use a hybrid approach, combining models to find the best intervention for the user for each stage in the program. Further research is required into how the models interact, and determine the feasibility of this idea.

3.4 Related research topics

In addition to the models discussed, we also need to consider the integration of a chatbot and the use of user feedback to improve our interventions.

To successfully achieve this, we need to perform more research into the effectiveness of chatbots; when to use them, and when not to.

Additionally, we want to look into how we can effectively learn to steer our interventions when the user provides feedback. We want to explore at least two paths: how to learn from indirect feedback, like swiping away or ignoring a notification, and how to learn from direct feedback, like a yes-or-no question we ask of the user.

4 Product Vision Board

The product vision board summarises what DETOUR is, who it is for, which problem it solves and why it is better than what already exists. It is the bridge between the Design Thinking phase and the next phase: every block on the board comes from one of the earlier phases, and it sets the scope for what we will build.

4.1 Product statement

> *DETOUR is a smartphone app for smokers who want to quit, built on existing models, like the HAPA behaviour change model. HAPA determines which phase of the quit process the user is in. Within that phase, COM-B and TDF identify the determinants that need to change, such as stress, triggering places or low motivation. Wearable data (heart rate/HRV), location, agenda and check-ins help measure these determinants. DETOUR then deploys the app features whose behaviour change techniques (BCTs) target exactly those determinants, and learns per user which ones work best.*

The statement is the short version of our How Might We question from the Define phase: *"How might we choose the correct (direct vs. indirect) kind of intervention (BCT) for the right person at the correct moment to stop smoking?"* It answers the three parts of that question:

- **Right person:** personal onboarding, plus learning per user over time.
- **Right moment:** knowing the phase (HAPA), plus live context from wearable, location, agenda and check-ins.
- **Right intervention:** BCTs chosen because they target the determinant behind *this* craving.

The behaviour models are not decoration. They are the logic the app runs on. The app works as a loop:

1. User info (onboarding, check-ins, sensors) is translated into **determinants** via TDF and COM-B, within the current **HAPA** phase.
2. The determinants select the **BCTs** that target them.
3. Each BCT can be delivered through several concrete **app features**. The app picks the feature that best fits this user.
4. A **performance evaluation** checks whether the feature helped: did the user interact, did the craving pass, did they smoke? The result feeds back into the selection, which is how the app gets more **personal** over time.

Table 1 shows this for one determinant. When the app detects that social influences matter for a user (for example, the user says in onboarding that friends or a partner support their attempt), it selects BCT 3.1 *Social support (unspecified)* from the BCT Taxonomy v1 (Michie et al., 2013). This BCT can be delivered through two different features. Which one the user gets depends on their profile, and the performance evaluation corrects the choice if it does not work.

Feature	**How it delivers social support**	**Suits users who…**	**Less suited for users who…**
Buddy system	The user links one trusted person (partner, friend, ex-smoker). When the user presses the SOS button during a craving, the buddy gets a notification and can call or send a message.	have one close person they want to involve; need help at the moment of a craving.	don't want others to know they are quitting; have no one to ask.
Leaderboard	The user joins a small group (friends or other DETOUR users) and sees each member's number of smoke-free days. The group can send each other encouragement.	are motivated by a group and by seeing progress; like a bit of competition.	feel ashamed when they fall behind; would be discouraged after a lapse.

Table 1: One BCT, two features. Determinant: social influences (TDF), which falls under social opportunity in COM-B. BCT: 3.1 Social support (unspecified). Each row is a separate feature that delivers the same BCT. The two right columns show which user profile the app uses to choose between them. If the chosen feature is ignored or does not reduce cravings, the performance evaluation lowers its score for that user and the other feature is tried.

4.2 Target group

Smokers who want to quit. This covers three groups that we met during the Empathize phase:

- people who tried before and failed,
- people who have stopped and want to keep it that way (relapse prevention),
- people who are trying for the first time.

All three share the condition we heard in every interview and in the first meeting with our challenger, Prof. Geert Crombez: **the motivation has to come from the user.** DETOUR does not try to convince people who don't want to quit; it helps people who already do.

4.3 Needs

We identified two core needs:

1. **Personal fit.** Smokers don't know which quit method or app suits them. There are many apps, most of low quality, and testing them one by one costs time and motivation. In practice, a failed attempt with the wrong app feels like a personal failure.
2. **Craving support that matches the cause.** Cravings have different causes (stress, a place, boredom, a social situation, physical withdrawal), and the same technique does not work for every cause. The HAPA phase also matters: someone still building motivation needs different help from someone resisting a craving on day three.

These needs come directly from the interviews (Empathize) and from the clustering of our HMW questions (Define), where "the right method for the right person" and "help at the moment of the craving" were the two largest clusters.

4.4 Product

- **Personal onboarding.** The user maps their own triggers and preferred kinds of help. Based on user feedback from the Test phase, this happens through a short chatbot conversation instead of a questionnaire.
- **Smart selection from a BCT library.** The app picks the techniques, not the user. Selection works on two levels:
  - **Phase (HAPA):** before the quit day, only motivation modules (health risks, money wasted, confidence). After the quit day, only coping modules (coping plans, trigger management, streaks).
  - **Cause (COM-B):** stress gets urge-surfing audio, a trigger place gets a problem-solving chatbot, low motivation gets the user's personal "why".

- **Several features per BCT.** The same technique can be delivered in different ways (Table 1), so the app can match the form to the person.
- **Learning per user.** Through the performance evaluation, the app learns which features work for this user and which ones they ignore.

4.5 Value

- **No need to compare and test apps yourself.** DETOUR picks the right techniques for you.
- **Help that fits the cause of this craving and the phase you're in**, so less wasted effort and a higher chance of quitting for good.

Users should not fail their quit attempt because the method didn't suit them.

4.6 Competitors and alternatives

Most quit apps (e.g. QuitNow, Kwit, Smokefree, Stoppen met Roken) do one of two things: they track progress and savings, or they offer the same fixed step-by-step programme to everyone. A few are science-based (Quit Genius, QuitSure, WHO Quit Tobacco), but they also follow a fixed path.

Apps that adapt to the user's context in real time exist only in research, for example Q Sense (geofence-triggered support), Smart-T (EMA-based lapse prediction) and Sense2Stop (stress detection from wearable sensors). These prove the idea is feasible, but none is on the market for consumers. DETOUR fills that gap by combining the scientific basis of these research apps with the usability of consumer apps. See the State of the Art chapter for the full comparison.

4.7 Link with the recommendations

The vision board describes *what* we will build. The recommendations from the Test phase set limits on *how* (Table 2).

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Recommendation	Effect on the product
Keep actions concrete	Every BCT module ends in one clear action, e.g. “do this 3-minute exercise”, not “try to relax”
Long questionnaires don’t work; a chatbot works better	Onboarding and check-ins are short chatbot conversations
Long chats are not pleasant either	Chatbot conversations are capped at a few turns
Showing the craving level can backfire	Risk is used internally to time interventions, but is not shown to the user as a score
Some users dislike daily notifications	Notification frequency is part of the personal profile and is learned over time
Adapt language to the profile	Tone (formal, informal, technical) is set during onboarding

Table 2: Recommendations from the Test phase and how they change the product described on the vision board.

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